User Manual

ISE Coursework — Bug Report Classification Tool

1 What This Tool Does

This tool runs two bug report classifiers side by side — a Naive Bayes + TF-IDF baseline and a Random Forest + Enhanced TF-IDF solution — and compares their performance on five deep learning project datasets. It runs each classifier 30 times with different train/test splits, reports Precision, Recall, and F1 for each, and runs Wilcoxon signed-rank tests to check whether any differences are statistically meaningful.

2 Folder Structure

```
ise-bug-classification/
├── classifier_fixed.py    the main script
├── data/                  put your CSV files here
│   ├── TensorFlow.csv
│   ├── PyTorch.csv
│   ├── Keras.csv
│   ├── MXNet.csv
│   └── Caffe.csv
├── results/               created automatically when you run
│   ├── summary.csv
│   ├── TensorFlow_baseline_raw.csv
│   ├── TensorFlow_solution_raw.csv
│   └── ... (one file per project per approach)
├── requirements.pdf
├── manual.pdf
└── replication.pdf
```

3 Running the Tool

Step 1 — Get the data

Go to `https://github.com/ideas-labo/ISE/tree/main/lab1` and download the five CSV files into your `data/` folder.

Step 2 — Install dependencies

```
pip install scikit-learn scipy numpy pandas
```

Step 3 — Run

```
python classifier_fixed.py
```

If your data is somewhere else:

```
python classifier_fixed.py --data_dir /your/path/to/data --output_dir ./results
```

4 Reading the Output

The script prints a table to the console for each project showing mean Precision, Recall, and F1 across 30 runs for both the baseline and the solution, along with the Wilcoxon p-value, A12 effect size, and whether the difference is statistically significant. A macro-average summary across all five projects is printed at the end.

```
  Project: TensorFlow
```

```
  ──────────────────────────────────────────────────────────────
  Metric       Baseline (mean±std)  Solution (mean±std)  p-val   A12  Sig?
  precision      1.000 ± 0.000        1.000 ± 0.000      1.0000  0.500   ✗
  recall         0.411 ± 0.132        1.000 ± 0.000      0.0000  1.000   ✓
  f1             0.571 ± 0.130        1.000 ± 0.000      0.0000  1.000   ✓
```

CSV files are also saved to the `results/` folder — one raw file per project per approach containing all 30 per-run scores, plus a summary CSV with means, standard deviations, p-values, and A12 scores for every metric and project.

5 Command-Line Options

```
--data_dir    path to folder containing CSVs   (default: ./data)
--output_dir  path to save results CSVs        (default: ./results)
--text_col    force a specific text column name   (optional override)
--label_col   force a specific label column name  (optional override)
```

6 Troubleshooting

FileNotFoundError

The CSV files need to be named exactly: TensorFlow.csv, PyTorch.csv, Keras.csv, MXNet.csv, Caffe.csv. Check the `data/` folder and make sure the names match.

ModuleNotFoundError

Run `pip install scikit-learn scipy numpy pandas` and try again.

Results look wrong

Run with `--text_col title_body --label_col label` to manually specify the column names if auto-detection is not picking them up correctly.